

An In Vivo High-Resolution Human Brain Atlas of Synaptic Density

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Synapses are fundamental to the function of the central nervous system and are implicated in a number of brain disorders. Despite their pivotal role, a comprehensive imaging resource detailing the distribution of synapses in the human brain has been lacking until now. Here, we employ high-resolution PET neuroimaging in healthy humans (17F/16M) to create a 3D atlas of the synaptic marker Synaptic Vesicle glycoprotein 2A (SV2A). Calibration to absolute density values (pmol/ml) was achieved by leveraging postmortem human brain autoradiography data. The atlas unveils distinctive cortical and subcortical gradients of synapse density that reflect functional topography and hierarchical order from core sensory to higher-order integrative areas—a distribution that diverges from SV2A mRNA patterns. Furthermore, we found a positive association between IQ and SV2A density in several higher-order cortical areas. This new resource will help advance our understanding of brain physiology and the pathogenesis of brain disorders, serving as a pivotal tool for future neuroscience research.

Key words: autoradiography; brain atlas; brain topography; neuroimaging; PET; positron emission tomography; SV2A; synaptic density

Significance Statement

Here, we present a high-resolution 3D in vivo brain atlas of synaptic density in the human brain. In the healthy human brain, distinctive cortical and subcortical gradients of synapse density reflect functional topography and hierarchical order from core sensory to higher-order integrative areas—a distribution that diverges from SV2A mRNA patterns. This brain atlas will help advance our understanding of human brain physiology and the pathogenesis of brain disorders, serving as a pivotal tool for future research in clinical, translational, and comparative neuroscience.

Introduction

Synapses are the junctions between neurons and constitute a central site for neuronal signaling in the brain. Estimates of the number of synapses formed on a single neuron in the adult human brain vary greatly (7,200–80,000), bringing their total number to a staggering

range of 6×10^{14} – 7×10^{15} (Silbereis et al., 2016). Because of their pivotal role in brain functions, the study of synapses is key to improving our understanding of how modulation of synapses affects brain function and human behavior from motor control to higher-order cognitive functions in both health and disease.

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Synaptic structures and their organization in the human brain can be studied in vitro with electron microscopy to visualize distinct components of individual synapses on a subcellular scale, with in situ hybridization for mRNA quantification, and with immunohistochemistry for visualizing specific synaptic proteins (Serrano et al., 2022). Commonly used human brain atlases that provide estimates of gene expression (e.g., mRNA quantification) have been shown to misrepresent the corresponding protein levels and distribution (Beliveau et al., 2017; Nørgaard et al., 2021; Hansen et al., 2022b). Autoradiography provides absolute target protein concentrations in postmortem brain tissue with high in-plane resolution, but the spatial information is often limited by a low number of discrete samples in a few subjects, and regions with microanatomical and functional heterogeneity are often represented by single estimates. To the best of our knowledge, no detailed human brain atlas that maps the synaptic density currently exists.

Using appropriate radioligands and large numbers of subjects, in vivo molecular neuroimaging with positron emission tomography (PET) can provide 3D human brain atlases at high resolution (Beliveau et al., 2017; Nørgaard et al., 2021; Hansen et al., 2022a). These PET-based brain atlases are instrumental in cross-modal analyses, revealing associations between spatial protein distributions and patterns of, e.g., cortical thickness and abnormalities in psychiatric and neurological disorders.

[¹¹C]UCB-J is a PET radioligand that binds to the Synaptic Vesicle glycoprotein 2A (SV2A), a transmembrane protein located in synaptic vesicles in the presynaptic axon terminals (Finnema et al., 2016; Nabulsi et al., 2016). SV2A is ubiquitously expressed in all gray matter regions of the brain and across neurotransmitter systems (Bajjalieh et al., 1994). SV2A colocalizes with synaptophysin immunostaining and is regarded as a proxy of synaptic density (Finnema et al., 2016; Rossi et al., 2022), with the majority of vesicles (97%) containing five molecules per vesicle (Mutch et al., 2011), allowing for investigations of how synaptic density relates to brain functions. For example, studies have shown that impaired cognition correlates negatively with PET measures of SV2A binding in Alzheimer's disease and schizophrenia (Mecca et al., 2020; Radhakrishnan et al., 2021; Zhang et al., 2023). A related question is to what extent synaptic density is functionally related to cognitive function in healthy individuals.

We here present a high-resolution in vivo human brain atlas of the synaptic marker SV2A. By obtaining quantitative measures from postmortem human brain autoradiography, we convert PET binding estimates to represent SV2A in absolute density (pmol/Lm brain tissue). The atlas represents synaptic density across the cerebral cortex and subcortical brain structures and enables us to characterize the spatial distribution of SV2A protein density and the relation to SV2A mRNA. Finally, we examine the relationship between synaptic density and demographic variables, brain morphology, and cognitive function.

Materials and Methods

Participants. Thirty-three healthy participants who had undergone [¹¹C]UCB-J PET and structural MRI examinations were included in the study. Demographic information, intelligence quotient (IQ) scores, and scan-related details are provided in Table 1. All participants were scanned under study protocols approved by the Ethics Committee of the Capital Region of Denmark (protocol ids: H-18038352, H-19053510, H-16026898, H-KF-2006-20) and data were entered into the CIMBI database (Knudsen et al., 2016).

All participants provided written informed consent before undergoing a comprehensive screening procedure involving medical history, blood biochemistry assessments, and neurological and psychiatric

Table 1. Demographics of the PET cohort (N = 33)

Sex (n, F/M)	17/16
Age (years)	24.0 [20, 44]
IQ ^a	108 [96, 125]
Injected dose (MBq)	410 [224, 550]
Injected dose, adjusted (MBq/kg)	5.5 [2.7, 9.7]
Injected mass, adjusted (ng/kg)	6.5 [1.2, 80.9]

Description of the sample used for the [¹¹C]UCB-J PET measurements. Values are median [range]. See Extended Data Table 1-1 for details on postmortem donors.

^aIQ scores were available for 30 participants.

evaluations. PET and MRI data from 12 participants have been reported in a previous publication (Johansen et al., 2023).

IQ was assessed with the Reynolds Intellectual Screening Test (RIST; Reynolds and Kamphaus, 2005) consisting of a verbal task (Guess What) capturing crystallized intelligence and a nonverbal task (Odd-Item Out) capturing fluid intelligence. Raw scores on the two tasks were combined and transformed into a total RIST index score reflecting overall IQ. Testing took place in standardized test room and was performed by trained neuropsychological testers.

PET and MRI acquisition. The detailed methodology for [¹¹C]UCB-J PET, T1- and T2-weighted MRI scan acquisitions, as well as blood analyses, can be found in a previous publication (Johansen et al., 2023). Briefly, participants underwent 120 min [¹¹C]UCB-J PET scans with the High-Resolution Research Tomograph (HRRT, CTI/Siemens) starting at the time of radiotracer bolus injection. Data were acquired in 3D list mode and reconstructed into 40 frames (8 × 15 s, 8 × 30 s, 4 × 60 s, 5 × 120 s, 10 × 300 s, 5 × 600 s) using a 3D OP-OSEM algorithm (16 subsets, 10 iterations) with attenuation correction using the HRRT maximum a posteriori transmission reconstruction method (MAP-TR) μ-map (Sureau et al., 2008; Keller et al., 2013). The reconstructed PET images comprised 207 planes of 256 × 256 voxels of 1.22 × 1.22 × 1.22 mm³. Continuous automatic arterial blood sampling was conducted for the first 15 min, followed by manual samples taken at specific time intervals (2.5, 5, 10, 25, 40, 60, 90, and 120 min). These samples were then used to determine metabolite-corrected arterial plasma input functions.

MRI scans were conducted using a Siemens Magnetom Prisma 3 T scanner (Siemens) using a Siemens 32-channel head coil. Participants underwent scanning with either of two similar T1 MP-RAGE sequences (TR, 2,000/1,810 ms; TE, 2.58 ms; TI, 972 ms; flip angle, 8°/9°; in-plane resolution, 0.9 × 0.9 mm²/0.8 × 0.8 mm²; slice thickness, 0.9 mm/0.8 mm; number of slices, 224; no gap; in-plane matrix, 288 × 288/256 × 256; field-of-view, 230 mm) and a T2-weighted space turbo spin echo sequence (TR, 3,200 ms; TE, 408 ms; in-plane resolution, 0.9 × 0.9 mm²; slice thickness, 0.9 mm; number of slices, 224; no gap; in-plane matrix, 256 × 256). T1- and T2-weighted images were corrected for gradient nonlinearities online on the scanner.

Image processing. The reconstructed dynamic PET data were motion corrected using the frame-by-frame *reconcile* procedure in AIR (v. 5.2.5, LONI, UCLA; <http://air.bmap.ucla.edu/AIR5>). Subsequent preprocessing of PET and MR images was performed using the PETSURFER pipeline (Greve et al., 2014) in FreeSurfer (Fischl, 2012; v. 7.2, <https://surfer.nmr.mgh.harvard.edu/>) as described in detail previously (Beliveau et al., 2017). The T1-weighted images underwent standard anatomical processing using the *recon-all* stream within FreeSurfer. PET images were coregistered to the MR images using a rigid transformation between the time-averaged PET image and the T1-weighted image and subsequently resampled onto the standard surface (*fsaverage*) and volume (*MNI152*) spaces and smoothed with 10 and 5 mm full-width at half-maximum, respectively. Regional time-activity curves for the regions of the Desikan-Killiany cortical atlas (Desikan et al., 2006), which subdivides the cortex into 34 gyral based regions of interest (the complete list of regions is available in Table 2), and eight subcortical regions defined by FreeSurfer (i.e., accumbens, amygdala, caudate, cerebellum cortex, hippocampus, putamen, thalamus, and ventral diencephalon) were extracted from the nonsmoothed data.

Kinetic modeling. Kinetic modeling of [^{11}C]UCB-J PET data was performed in R (v. 4.2.1, R Foundation) using the *kinfitr* package v. 0.7.2 (Tjerkaski et al., 2020). The metabolite-corrected arterial input function (AIF) was estimated for each participant by first fitting a cubic spline to the first four samples of the whole blood-to-plasma ratio data. Interpolated values were then multiplied with the continuous whole blood curve and subsequently concatenated with the manual plasma samples. The unmetabolized [^{11}C]UCB-J in plasma (parent fraction) obtained from the radio-HPLC analysis was fitted using a Hill model, which was subsequently multiplied to the plasma radioactivity curve to obtain the metabolite-corrected AIF.

Total tissue distribution volumes (V_T ; mL/cm 3) of all brain regions, as well as for the cortical vertices and subcortical voxels, were estimated with the one-tissue compartment model (1TCM) using the participants' metabolite-corrected AIF, as previously described (Johansen et al., 2023).

In vivo PET SV2A binding and in vitro autoradiography. To obtain regional estimates of cerebral SV2A protein densities (B_{max} ; pmol/mL), we made use of frozen sections of postmortem human brain tissue from seven donors (6F/1M, median age [range] 88 [68–91] years) from the Bispebjerg Brain Bank (Bispebjerg-Frederiksberg Hospital, Copenhagen University Hospital), approved by the Ethics Committee of the Capital Region of Denmark (protocol id: H-20062005). The sections included frontal, parietal, temporal, and occipital cortex as well as white matter. We employed [^3H]UCB-J in vitro autoradiography to determine SV2A B_{max} . Frozen brain pieces ($\sim 4\text{--}5\text{ cm}^3$) from the frontal, temporal, parietal, and occipital cortices containing both gray and white matter were dissected and stored at -80°C . Coronal cryosections (12 μm) were prepared. In vitro autoradiography was performed as described in Raval et al. (2021).

A saturation assay was performed on a subset of frontal cortex brain slices using varying concentrations of [^3H]UCB-J to determine total binding and [^3H]UCB-J combined with 10 mM levetiracetam (Keppra, UCB Pharma) to assess nonspecific binding (NSB). The B_{max} of [^3H]UCB-J was determined to be 497 fmol/mg TE (95% CI = 466–530 fmol/mg TE) and the K_D to be 3.2 nM (95% CI = 2.6–3.8 nM). Since the levetiracetam-blocked sections yielded similar results as the unblocked white matter sections, white matter was used as a measure of NSB. For regional estimation of SV2A B_{max} , all cortical sections were incubated in 20 nM [^3H]UCB-J ($\sim 6\text{--}7$ times the K_D). These data were converted from the unit pmol/g to pmol/mL using a gray matter density of 1.045 g/mL (DiResta et al., 1991). To supplement the data with subcortical regions, we included SV2A estimates from [^{11}C]UCB-J in vitro autoradiography conducted by Varnäs et al. (2020). These data originated from three donors (1F/2M, median age [range] 34 [32–62] years, 2 slices per donor) and were reported in the unit of kBq/mm 2 . To align the estimates with our [^3H]UCB-J data, we performed a linear regression with a null intercept to obtain estimates in units of pmol/mL.

Next, weighted averages of the in vivo [^{11}C]UCB-J V_T estimates corresponding to the autoradiography regions were obtained, and the association with the calibrated autoradiography data was assessed using linear regression. The resulting linear equation was then used to transform the in vivo [^{11}C]UCB-J V_T values for all regions, voxels, and vertices into absolute protein densities (pmol/mL). All subsequent references to SV2A densities thus pertain to in vivo [^{11}C]UCB-J V_T values obtained with PET calibrated to pmol/mL using this approach.

All demographic information on the donors is summarized in Extended Data Table 1-1.

In vivo SV2A protein density and SV2A gene expression. As a measure of gene expression, regional in vitro SV2A mRNA data were obtained from six donors using the *abagen* toolbox (v 0.1.3; Markello et al., 2021). This software provides a standardized pipeline for the quantification of mRNA data from the Allen Human Brain Atlas (Hawrylycz et al., 2012).

Statistics. For each region, a linear regression model was used to assess the association between age, sex, and IQ and in vivo SV2A densities. We further assess the effect of cortical thickness and subcortical volume on in vivo SV2A densities, adjusting for age and sex (and intracranial volume

for voxel-based analyses). All tests were corrected for multiple comparisons using the false discovery rate method (Benjamini and Hochberg, 1995). Regional in vitro SV2A mRNA levels and in vivo SV2A densities were compared using a Spearman correlation.

Additional exploratory analyses were performed at the vertex and voxel level to investigate subregional associations between in vivo SV2A density and IQ using a generalized linear model adjusted for age and sex. Model significance was assessed with permutation testing using FreeSurfer's *fsPALM* script, which invokes FSL's *palm* (alpha111; Winkler et al., 2014). Cluster-forming threshold was set to $p = 0.001$, cluster-wise p value to 0.05, and 5,000 permutations were performed. Bonferroni's correction for three spaces, accounting for both cortical hemispheres and the subcortical space, was applied.

Results

In vitro autoradiography and in vivo SV2A imaging

In our in vitro [^3H]UCB-J autoradiography experiments (Extended Data Fig. 1-1), we found high SV2A binding in the neocortical gray matter (range 368–629 pmol/g) and low levels in the white matter (27–36 pmol/g). The ratio between these two regions was similar to the corresponding regional values measured in Bq/mm 2 from Varnäs et al. (2020). We could thus convert these to pmol/g based on the linear regression equation (Extended Data Fig. 1-2; conversion factor = 0.71 pmol/g per kBq/mm 2). Finally, all estimates were converted to volume densities (pmol/mL).

The in vivo regional average [^{11}C]UCB-J V_T estimates correlated well with the postmortem in vitro autoradiography data (Fig. 1; $r_{(11)} = 0.85$; $p = 0.0002$) and were described by the following equation: $V_T = 3.56\text{ mL/cm}^3 + 0.03\text{ mL/cm}^3/\text{pmol/mL} \times B_{\text{max}}$ (pmol/mL). Here, the intercept estimate [SE] of 3.56 [2.18] mL/cm 3 represents the average nondisplaceable compartment of PET [^{11}C]

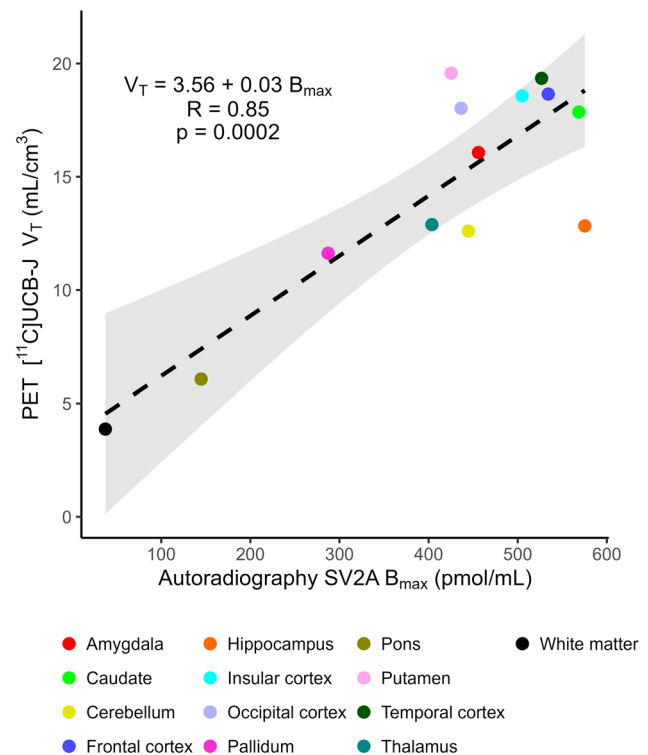


Figure 1. Average regional SV2A densities measured with in vivo [^{11}C]UCB-J PET (V_T ; mL/cm 3) and in vitro autoradiography (B_{max} ; pmol/mL) in brains from healthy participants/donors. The dashed line represents the regression line and shaded gray area the 95% CI. See Extended Data Figures 1-1 and 1-2 for more details on the autoradiography quantification.

UCB-J binding (V_{ND}), while the slope estimate [SE] of 0.03 [0.005] is equal to the average in vivo [^{11}C]UCB-J plasma free fraction (f_P) divided by the in vivo [^{11}C]UCB-J dissociation constant (f_P/K_D).

The in vivo synaptic density atlas

The linear regression equation from Figure 1 was used to calibrate the V_T values and thereby create the in vivo cortical (vertex-based) and subcortical (voxel-based) SV2A density maps (Fig. 2). The atlas shows high in vivo SV2A density in the neocortex, with highest densities in the posterior cingulate, precuneus, and gyri of the temporal cortex and lowest in the pre- and postcentral gyri. For the subcortical regions, we observe the highest density in the putamen and caudate and lower density in the hippocampus, thalamus, pallidum, and cerebellum. Regional in vivo SV2A densities are reported in Table 2.

We also created maps visualizing the variability of the in vivo SV2A density (Extended Data Figs. 2-1, 2-2): Relative to the mean value, we observe the highest cortical variability bilaterally in the pericollateral rim of the cingulate cortex, around the

temporal poles and inferior temporal gyri, medial post-central gyrus, and part of the left lateral occipital cortex. In contrast, the precuneus bilaterally and frontal and temporal areas of the right hemisphere show the lowest variability. By visual inspection, the variability appears higher in the left hemisphere.

In vivo SV2A density and in vitro SV2A gene expression

We examined how well regional in vivo SV2A density corresponded with SV2A gene expression as assessed by in vitro SV2A mRNA data from the Allen Human Brain Atlas (Hawrylycz et al., 2012). We found no association between regional in vivo SV2A density and in vitro SV2A mRNA levels (Fig. 3; Spearman's $r_{(39)} = 0.10$; $p = 0.53$).

Effects of demographic variables on in vivo SV2A density

Across all brain regions, we found no effects of age, sex on in vivo SV2A densities, or association with IQ scores when correcting for multiple comparisons (all corrected p values >0.13). Similarly, there were no associations between cortical thickness, subcortical volumes, and regional in vivo SV2A densities (all corrected p values

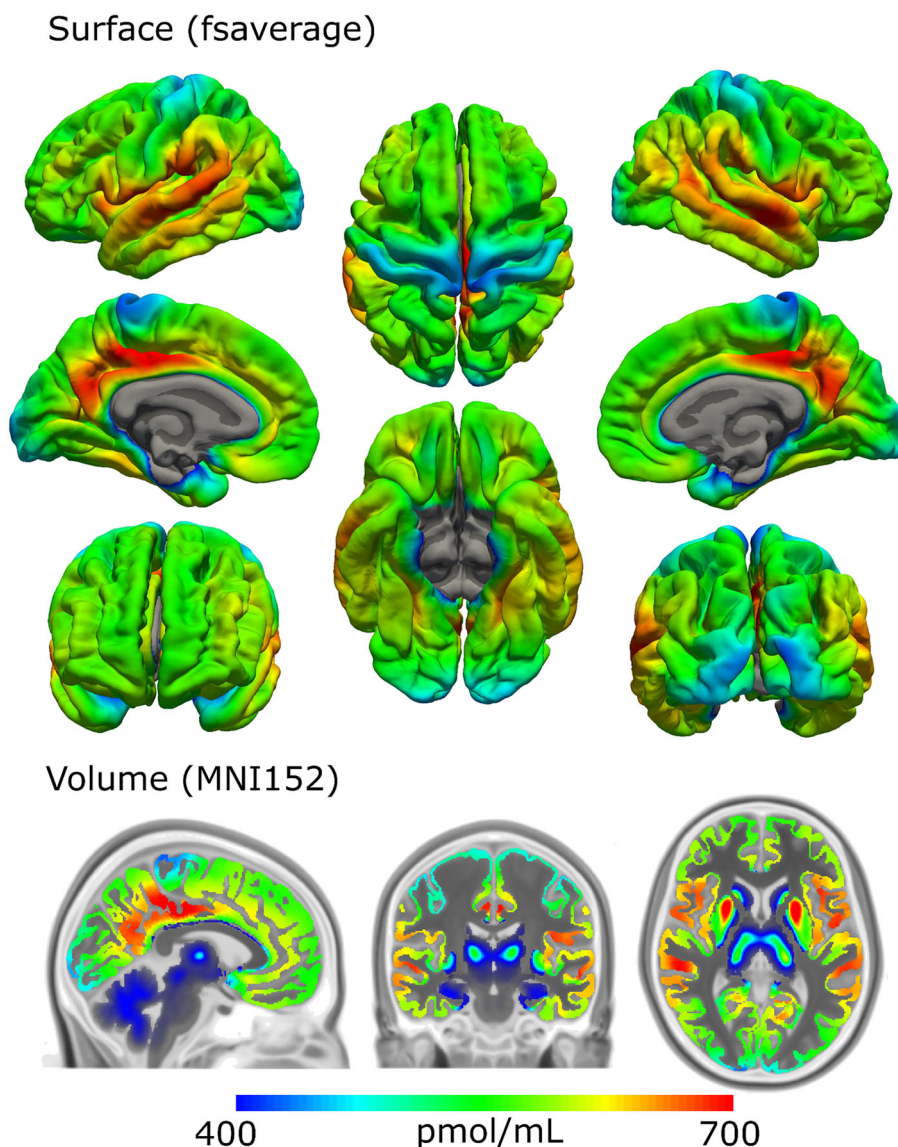


Figure 2. Average in vivo SV2A density maps (B_{max} ; pmol/mL) on the FreeSurfer *fsaverage* surface (top) and in the *MNI152* volume space (bottom). See Extended Data Figures 2-1–2-4 for more details.

Table 2. Summary of SV2A $B_{\max}$ (pmol/mL) in individual regions in the left and right hemisphere and for a weighted average (total)

Lobe	Region	Total			Left hemisphere			Right hemisphere			Left/right ratio		
		Mean	SD	COV (%)	Mean	SD	COV (%)	Mean	SD	COV (%)	Mean	SD	COV (%)
Cingulate	Caudal anterior cingulate	526	89	16.92	539	97	18.00	513	85	16.57	1.05	0.09	8.57
	Isthmus cingulate	484	78	16.12	483	76	15.73	485	82	16.91	1.00	0.06	6.00
	Posterior cingulate	577	89	15.42	583	95	16.30	572	85	14.86	1.02	0.06	5.88
Frontal	Rostral anterior cingulate	565	86	15.22	572	87	15.21	555	88	15.86	1.03	0.06	5.83
	Caudal middle frontal	564	84	14.89	574	87	15.16	554	81	14.62	1.04	0.03	2.88
	Frontal (total)	569	83	14.59	573	86	15.01	565	81	14.34	1.01	0.03	2.97
	Frontal pole	570	100	17.54	560	107	19.11	578	98	16.96	0.97	0.06	6.19
	Lateral orbitofrontal	564	87	15.43	574	93	16.20	555	84	15.14	1.03	0.05	4.85
	Medial orbitofrontal	559	86	15.38	558	88	15.77	560	87	15.54	1.00	0.05	5.00
	Paracentral	534	84	15.73	536	86	16.04	533	83	15.57	1.01	0.04	3.96
	Pars opercularis	625	93	14.88	626	99	15.81	624	88	14.10	1.00	0.05	5.00
	Pars orbitalis	578	88	15.22	568	97	17.08	586	86	14.68	0.97	0.08	8.25
	Pars triangularis	594	85	14.31	595	92	15.46	592	83	14.02	1.01	0.06	5.94
	Precentral	547	80	14.63	553	83	15.01	541	78	14.42	1.02	0.04	3.92
	Rostral middle frontal	580	86	14.83	582	92	15.81	579	83	14.34	1.00	0.05	5.00
	Superior frontal	569	84	14.76	574	86	14.98	564	83	14.72	1.02	0.02	1.96
Occipital	Cuneus	572	85	14.86	571	88	15.41	574	84	14.63	0.99	0.04	4.04
	Lateral occipital	520	81	15.58	510	87	17.06	530	79	14.91	0.96	0.07	7.29
	Lingual	560	82	14.64	563	85	15.10	557	80	14.36	1.01	0.03	2.97
	Occipital (total)	546	82	15.02	540	85	15.74	551	80	14.52	0.98	0.04	4.08
	Pericalcarine	574	87	15.16	562	85	15.12	584	91	15.58	0.96	0.03	3.12
Parietal	Inferior parietal	598	84	14.05	587	87	14.82	607	86	14.17	0.97	0.06	6.19
	Parietal (total)	577	85	14.73	576	88	15.28	578	83	14.36	1.00	0.04	4.00
	Postcentral	524	80	15.27	527	86	16.32	521	76	14.59	1.01	0.04	3.96
	Precuneus	616	88	14.29	616	90	14.61	615	87	14.15	1.00	0.02	2.00
	Superior parietal	542	84	15.50	544	85	15.62	541	84	15.53	1.01	0.03	2.97
	Supramarginal	613	92	15.01	618	97	15.70	609	90	14.78	1.02	0.06	5.88
Temporal	Banks of the superior temporal sulcus	631	89	14.10	632	93	14.72	631	91	14.42	1.00	0.07	7.00
	Entorhinal	491	102	20.77	502	110	21.91	479	96	20.04	1.05	0.10	9.52
	Fusiform	577	91	15.77	582	95	16.32	573	88	15.36	1.01	0.04	3.96
	Inferior temporal	594	98	16.50	598	106	17.73	590	92	15.59	1.01	0.05	4.95
	Insula	566	81	14.31	568	83	14.61	565	82	14.51	1.01	0.05	4.95
	Middle temporal	622	90	14.47	618	97	15.70	625	86	13.76	0.99	0.06	6.06
	Parahippocampal	541	98	18.11	538	104	19.33	545	98	17.98	0.99	0.09	9.09
	Superior temporal	606	89	14.69	603	95	15.75	609	84	13.79	0.99	0.05	5.05
	Temporal (total)	595	91	15.29	596	97	16.28	595	87	14.62	1.00	0.04	4.00
	Temporal pole	529	105	19.85	541	111	20.52	517	106	20.50	1.05	0.10	9.52
Subcortical	Transverse temporal	600	85	14.17	607	91	14.99	591	84	14.21	1.03	0.07	6.80
	Accumbens	598	108	18.06	616	121	19.64	585	103	17.61	1.05	0.09	8.57
	Amygdala	472	84	17.80	471	89	18.90	474	82	17.30	0.99	0.05	5.05
	Caudate	539	81	15.03	548	81	14.78	531	83	15.63	1.04	0.05	4.81
	Cerebellum	341	55	16.13	347	58	16.71	335	54	16.12	1.04	0.04	3.85
	Hippocampus	350	64	18.29	351	67	19.09	348	62	17.82	1.01	0.05	4.95
	Pallidum	304	52	17.11	315	59	18.73	294	51	17.35	1.08	0.12	11.11
	Pons	95	30	31.58									
	Putamen	604	88	14.57	618	92	14.89	591	85	14.38	1.05	0.03	2.86
	Thalamus	352	55	15.62	353	55	15.58	351	57	16.24	1.01	0.07	6.93
	Ventral dc	170	37	21.76	173	39	22.54	167	37	22.16	1.04	0.10	9.62
	White matter	12	19	158.33									

See Extended Data Tables 2-1–2-3 for more details.

>0.69). All model outputs are listed in Extended Data Tables 2-1 and 2-2.

Our vertex- and voxel-wise analysis showed positive associations between IQ scores and in vivo SV2A density in several symmetrical vertex clusters found in the parahippocampal gyri and orbitofrontal and temporal cortices (Fig. 4 and Extended Data Table 2-3). No subcortical voxels were below the set threshold.

Discussion

Here, we present the first high-resolution in vivo human brain atlas of absolute SV2A density as a proxy for synaptic density.

We combined in vivo and in vitro molecular imaging data of SV2A to create a brain map using an optimized surface- and volume-based approach. The atlas provides the spatial distribution and patterns of synaptic density in the human neocortex and subcortical brain structures and is freely available for easy comparison with other maps in *fsaverage* space, while the subcortical structures are available in the commonly used *MNI152* template.

Whereas the regional PET estimates of in vivo SV2A binding generally correlated very well with postmortem in vitro autoradiography measurements of SV2A density, some regions, and particularly the hippocampus, exhibited notable differences between

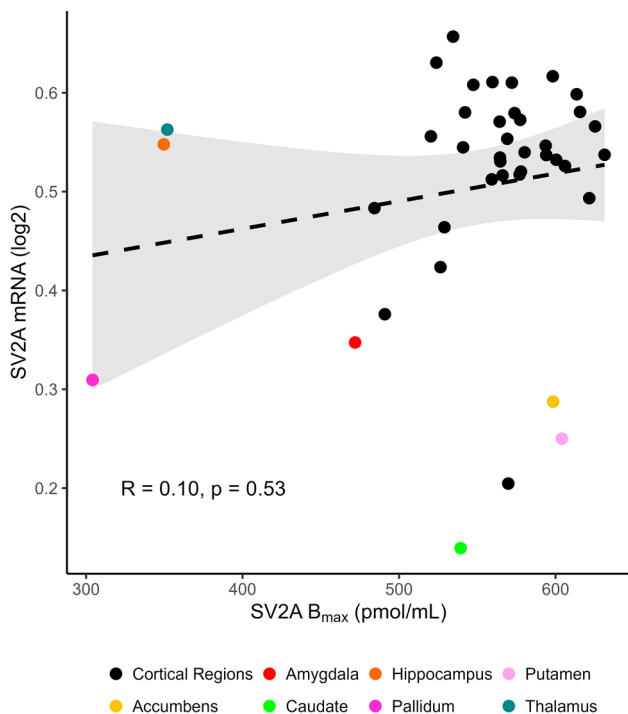


Figure 3. Average regional in vivo SV2A density plotted against corresponding in vitro SV2A mRNA values ($\log_2$). The dashed line and shaded gray area represent the regression line and 95% CI for all brain regions. SV2A density and mRNA levels were not correlated, as evaluated with Spearman's correlation.

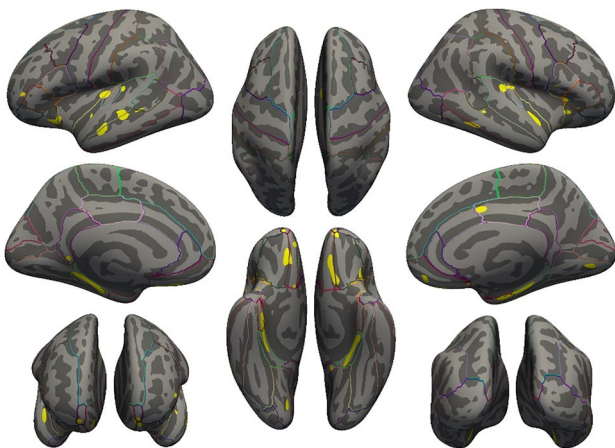


Figure 4. Inflated surface maps with projections of vertices that show positive association between in vivo SV2A density and IQ scores ($N = 30$). For region annotations, cluster sizes and exact p values, see Extended Data Table 2–3. The colored lines show the borders of the regions defined by the Desikan–Killiany atlas.

the autoradiography and PET data. In our data and in previous SV2A PET studies (Finnema et al., 2017; Lu et al., 2021; Naganawa et al., 2021), the hippocampus consistently ranks as a low-binding region whereas in the Varnäs et al. autoradiography data, the hippocampus is reported to have the highest SV2A density. Although partial volume effects leading to underestimation of PET measurements could potentially contribute to this finding, it seems unlikely that such effects would differentially influence the hippocampus estimates in this young and healthy cohort. An important consideration is that each PET measure represents the average of each region and that the

hippocampus is not segmented into gray and white matter, which may well dilute the specific SV2A signal somewhat. In contrast, the higher spatial resolution of autoradiography allows for better distinction between gray and white matter. However, autoradiography was performed on two discrete brain sections not covering entire regions and thus is more susceptible to outliers. In the Varnäs data, the hippocampus showed the largest variability, and given the diverse cytoarchitecture and neuroreceptor composition of the different hippocampal subfields (Palomero-Gallagher et al., 2020), the autoradiography samples may not fully capture the heterogeneity present in this brain region. Of note, when the hippocampus data point is omitted in the linear regression model, the estimates for the slope and intercepts are still within the 95% CI of the estimates based on all regions.

Conversely, most SV2A PET studies find the putamen to have the highest binding (Finnema et al., 2016; Naganawa et al., 2021), but this region only exhibited intermediate density in the autoradiography data. Interestingly, our atlas reveals a clear rostral-caudal gradient in the putamen. Functionally, the rostral part of the putamen, where we observe the highest binding, receives input from associative cortex in the frontal, parietal, and temporal lobes (Parent and Hazrati, 1995; Draganski et al., 2008) and contains almost twice as many interneurons as the caudal part (Lecumberri et al., 2018). The Varnäs autoradiography study sampled the caudal parts of the putamen, which receive projections from the sensorimotor cortex where we observe lower in vivo binding. Such a gradient could indicate that the striatal networks involved in motor function are less complex or require less modulation compared with the more rostral networks involved in higher-order cognitive functions.

In the neocortex, we observe the highest in vivo SV2A density in the posterior cingulate cortex, precuneus cortex, insular cortex, and temporal cortex. In contrast, we observe the lowest density in the lateral occipital cortices and the sensorimotor cortices, which include central parts of the primary visual cortex and the primary motor and sensory cortices. This pattern of higher density in regions related to associative functions and lower density in core sensory and motor areas aligns with patterns of many neurotransmitter receptors (Beliveau et al., 2017; Zilles and Palomero-Gallagher, 2017). Additionally, we observe a gradient within the sensorimotor cortex that may reflect the somatotopic homunculus (Penfield and Boldrey, 1937; Gordon et al., 2023): We see the lowest in vivo SV2A density in areas representing larger body structures, such as the trunk and legs, gradually increasing laterally to regions representing fine motor skills and higher sensory resolution, such as the hands, face, and mouth (Extended Data Fig. 2–3). Midlaterally in the central sulcus, between these two bands of lower SV2A density, we see a small island with slightly higher binding, which seems to overlap topographically with the newly discovered *Rolandic motor association area* (Jensen et al., 2023). In the occipital cortex, we likewise observe a gradient in SV2A density going from lower density in the occipital pole (central vision) to higher density in an anterior direction along the calcarine sulcus (peripheral vision). These spatial differences could imply that basic sensory and motor functions rely on simpler cortical networks, whereas higher-order integrative, cognitive, and emotional functions require more complex cortical networks with higher synaptic density. These hypotheses deserve to be investigated in future studies, e.g., combined with activation paradigms and functional MRI.

The hierarchical distinctions observed in in vivo SV2A distribution is also present for several neurotransmitter systems in the human and the macaque cortex (Palomero-Gallagher et al.,

2020; Goulas et al., 2021; Froudust-Walsh et al., 2023) and is further mirrored in a study by Wei et al. (2019) demonstrating that higher-order cognitive networks (e.g., the default mode network) show higher expression of the so-called *human-accelerated region* (HAR) genes compared with the somatomotor and visual networks, and that HAR genes are associated with, e.g., human intellectual ability. Whether higher synaptic density in these or other regions is associated with better cognitive ability in healthy individuals, as has been demonstrated for patients with neurodegenerative disorders (Mecca et al., 2022), is unclear. Interestingly, our exploratory vertex-based analysis showed positive associations between IQ scores and in vivo SV2A density in areas such as the parahippocampal gyrus, entorhinal cortex, and orbitofrontal and temporal cortices. Although this association warrants replication in a larger sample, characteristics of several of these regions have previously been linked to intelligence, by, e.g., a shared genetic origin with gray matter density in the parahippocampal gyrus (Pol et al., 2006), positive associations with cortical thickness, dendritic length and complexity, and speed of action potentials in the temporal cortex (Goriounova et al., 2018). In contrast, another study found no associations between IQ and absolute number of neurons, cortical thickness, or cortical surface area in the global neocortex (Songthawornpong et al., 2020). Together, this underscores the neurobiological complexity underlying cognitive abilities and that SV2A neuroimaging may prove a valuable tool for disentangling the relationship between synaptic density in distinct brain areas and different cognitive domains.

We found no spatial correlation between the regional in vivo SV2A density and in vitro SV2A mRNA from the Allen Human Brain Atlas. The absence of a correlation between SV2A gene expression (mRNA) and SV2A protein density is in agreement with SV2A being a presynaptic protein, as also seen for, e.g., the serotonin transporter and several neurotransmitter receptors (Beliveau et al., 2017; Hansen et al., 2022b). This discrepancy is explained by several factors, such as the proportion of projection neurons versus interneurons, stability of the mRNA, SV2A protein turnover, and capacity for local protein synthesis in the axon terminal (Hafner et al., 2019). In addition to being protein-based, the atlas presented here has the advantage that SV2A is expressed as absolute protein density (pmol/mL), thus making it directly comparable to other microanatomical structures such as the density of neurons and neuroreceptors. Further, we used surface-based processing which greatly reduces artifacts due to partial volume effects (Greve et al., 2014). This is especially evident when comparing to a volume-based population-average map of [^{11}C]UCB-J nondisplaceable binding potential (BP_{ND} ; Markello et al., 2022). The BP_{ND} map shows cortical striping patterns and a significant correlation with cortical curvature indicative of partial-volume effects which was not the case for our SV2A density map (Extended data Fig. 2-4).

This work is not without limitations. One aspect is the estimated parameters of the linear equations used to transform our in vivo PET measurements into absolute protein densities. For example, if we choose the most extreme values within the 95% confidence interval to provide the slope and intercept, the estimate of SV2A density for a cortical region with [^{11}C]UCB-J V_T value of $\sim 18 \text{ mL/cm}^3$ would change by $\pm 9\%$, whereas a lower binding region like the pallidum with a V_T of $\sim 12 \text{ mL/cm}^3$ would change within the range of -30% to $+7\%$. Having more low and intermediate SV2A binding regions would reduce the 95% CI of our slope estimates.

Another consideration is that the postmortem brain tissue came from donors who were older than the healthy individuals

in our PET cohort. We do not find any age effect on in vivo SV2A density but we cannot exclude that this is because of the relatively narrow age range. Some PET studies with broader age ranges have reported no age effects (Michiels et al., 2021; Andersen et al., 2022), while other studies suggest there may be a reduction with age (Mansur et al., 2020; Toyonaga et al., 2023). In a recent comprehensive study of 80 healthy individuals with a large and representative age range, Toyonaga et al. (2023) investigated age-related loss in brain volume and in SV2A as measured with [^{11}C]UCB-J-PET. They found age to explain only 9% of the variation in total gray matter specific SV2A binding, with an estimated reduction of 2.5% over a decade compared with the mean age of the sample. Different analysis approaches (e.g., [^{11}C]UCB-J quantification method, application of partial volume correction, and whether correction for multiple comparisons was taken into account) could explain the divergent results (Greve et al., 2016; Nørgaard et al., 2019). Furthermore, potential age effects may not follow a linear trajectory but could, for example, reflect the effects of synapse formation and pruning in late adolescence and young adulthood and a reduction after a certain age. As such, effects of age may best be explored with models informed by prior knowledge on brain development. However, depending on the age group to which the atlas is applied in future studies, a correction factor can be applied based on the Toyonaga dataset.

The participants from the in vitro autoradiography and the in vivo PET studies were not perfectly balanced in terms of sex, but in line with other SV2A PET studies (Fang et al., 2021; Michiels et al., 2021; Andersen et al., 2022), we do not find any effect of sex on cerebral SV2A densities.

Because the present cohort had relatively high IQ scores and limited variability, the participants in this study are not fully representative of the general population. It is possible that inclusion of participants with a larger range of IQ could have strengthened our finding of a positive association between IQ and SV2A, but this remains to be seen in future studies.

Finally, it is important to consider the limitations of using SV2A as a proxy for synaptic density. SV2A is a highly preserved synaptic protein alike synaptophysin which is one of the most commonly used synaptic density markers (Rossi et al., 2022), and measurements of the two proteins correlate very well (Finnema et al., 2016; Mikkelsen et al., 2023). SV2A is located in the synaptic vesicle membrane with most vesicles estimated to contain five molecules per vesicle (Mutch et al., 2011). Consequently, both PET and autoradiography imaging can theoretically be affected by variability in the number of vesicles per synapse or the number of proteins in each vesicle. These potential sources of variability are important to keep in mind as their contribution to different pathological conditions could vary.

Conclusion

Here, we present a detailed PET- and MRI-based high-resolution in vivo brain atlas of absolute synaptic density in the healthy human brain. By combining two radioligand neuroimaging techniques, we calibrate our in vivo PET measurements of SV2A to represent absolute protein densities. The atlas highlights distinctive cortical and subcortical gradients of synapse density that reflect functional topography and mirrors hierarchical patterns seen for cortical thickness and functional networks. Furthermore, our data indicates a potential link between general intelligence and synaptic density in several brain regions that deserved further exploration in future studies. This new resource will help advance our understanding of brain physiology and the pathogenesis of brain disorders, serving as a pivotal tool for future neuroscience research.

Data Availability

The atlas will be freely available and can be accessed from the site <https://xtra.nru.dk/SV2A-atlas>. Researchers may apply for access to the original data by submitting an application to the CIMBI Database (<https://cimbi.dk>; Knudsen et al., 2016). Code is available upon reasonable request.

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